Linear Algebra and Its Applications

Section 2.4: Partitioned Matrices

Detailed worked solutions

Mathjock

Text and numbering. David C. Lay, *Linear Algebra and Its Applications*, fourth edition (Pearson, 2012), as supplied in the textbook PDF linked on Mathjock. Labels give chapter, section, and exercise number; for example, 1.2.3 is Exercise 3 in Section 1.2. A P label denotes a practice problem, and S denotes a chapter supplementary exercise. These are independently written solutions, not a publisher's solutions manual. Short descriptions identify the problems; consult the textbook for the full original statements.

Source scope. The supplied textbook contains printed Chapters 1–8. Its separately published online Chapters 9–10 are absent from that PDF and are not included in this collection.

How to read the solutions. Start with the section's definitions. For a calculation, follow the row operations and substitute the result into the original equations. For a proof, identify the assumptions, the desired conclusion, and why each implication follows. Free variables are parameters, not values that should be silently set to zero. Unless otherwise specified, scalars are real; complex arithmetic is used where the problem requires it.

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2.4 Partitioned matrices

A block is itself a matrix. Block addition and multiplication follow the ordinary row–column rules, but every block product must have matching inner dimensions. Preserve the order of factors: AB cannot in general be changed to BA. An identity block has the size needed in its own row and column, so two symbols I in one display may have different sizes.

Practice problem 2.4.P1: Invert a block row-replacement matrix

Let $T = \begin{bmatrix} I & 0 \\ A & I \end{bmatrix}$. Try reversing the addition of A times the first block row to the second:

$$S = \begin{bmatrix} I & 0 \\ -A & I \end{bmatrix}.$$

Block multiplication gives $TS = \begin{bmatrix} I & 0 \\ A - A & I \end{bmatrix} = I$. In the reverse order ST has lower-left block $-A + A = 0$ and all other identity blocks. Thus $S = T^{-1}$.

Practice problem 2.4.P2: Transpose a horizontal partition before multiplying

For $X = [X_1 \ X_2]$, its transpose is vertical:

$$X^T X = \begin{bmatrix} X_1^T \\ X_2^T \end{bmatrix} [X_1 \ X_2] = \begin{bmatrix} X_1^T X_1 & X_1^T X_2 \\ X_2^T X_1 & X_2^T X_2 \end{bmatrix}.$$

The off-diagonal blocks do not disappear; they measure cross-products between the two groups of columns.

Exercise 2.4.1: Add a multiple of one block row

The conformability assumption makes each displayed product valid.

$$\begin{bmatrix} I & 0 \\ E & I \end{bmatrix} \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} A & B \\ EA + C & EB + D \end{bmatrix}$$

The lower-left block is $EA + IC = EA + C$; the upper row is unchanged.

Exercise 2.4.2: Multiply by a block diagonal matrix

The conformability assumption makes each displayed product valid.

$$\begin{bmatrix} E & 0 \\ 0 & F \end{bmatrix} \begin{bmatrix} P & Q \\ R & S \end{bmatrix} = \begin{bmatrix} EP & EQ \\ FR & FS \end{bmatrix}$$

The first block row is multiplied by E and the second by F; the zero blocks remove the other terms.

Exercise 2.4.3: Interchange two block rows

The conformability assumption makes each displayed product valid.

$$\begin{bmatrix} 0 & I \\ I & 0 \end{bmatrix} \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} C & D \\ A & B \end{bmatrix}$$

For example the upper-left product is $0A + IC = C$. This is a block row interchange.

Exercise 2.4.4: Subtract a block multiple

The conformability assumption makes each displayed product valid.

$$\begin{bmatrix} I & 0 \\ -E & I \end{bmatrix} \begin{bmatrix} W & X \\ Y & Z \end{bmatrix} = \begin{bmatrix} W & X \\ Y - EW & Z - EX \end{bmatrix}$$

The minus sign acts on both products in the lower block row. The order EW, rather than WE, comes from the row–column rule.

Exercise 2.4.5: Equate the four blocks before choosing inverses

Multiplication of the two given matrices gives

$$\begin{bmatrix} A + BX & BY \\ C & 0 \end{bmatrix} = \begin{bmatrix} 0 & I \\ Z & 0 \end{bmatrix}.$$

Thus the exact requirements are $BX = -A$, $BY = I$, and $Z = C$. If B is square, these requirements force B invertible, and

$$X = -B^{-1}A, \qquad Y = B^{-1}, \qquad Z = C.$$

Indeed $B(-B^{-1}A) = -A$ and $BB^{-1} = I$, checking both upper blocks. More generally, if B is rectangular and has a right inverse Y, the choice $X = -YA$ still works. Without such a right inverse, $BY = I$ is impossible, so no solution exists.

Exercise 2.4.6: Invert a block lower triangular matrix

The product is

$$\begin{bmatrix} XA & 0 \\ YA + ZB & ZC \end{bmatrix} = I.$$

Assume A and C are square and invertible. Then the diagonal blocks give $X = A^{-1}$ and $Z = C^{-1}$. The lower-left equation is $YA = -C^{-1}B$, so right multiplication by A^{-1} gives

$$Y = -C^{-1}BA^{-1}.$$

Inserting these formulas makes $YA + ZB = -C^{-1}B + C^{-1}B = 0$. For square diagonal blocks, their invertibility is also necessary because $XA = I$ and $ZC = I$ imply it.

Exercise 2.4.7: A zero middle block does not contribute

The block product is

$$\begin{bmatrix} XA & XZ \\ YA+B & YZ+I \end{bmatrix} = I.$$

With A square and invertible, $XA = I$ gives $X = A^{-1}$. Then $XZ = 0$ forces $Z = 0$, and $YA + B = 0$ gives $Y = -BA^{-1}$. The remaining equation is $YZ + I = I$, which is now satisfied.

Conclusion. $X = A^{-1}$, $Y = -BA^{-1}$, and $Z = 0$ under the stated square-invertible assumption.

Exercise 2.4.8: Match three block columns

Multiplication gives

$$\begin{bmatrix} AX & AY & AZ+B \\ 0 & 0 & I \end{bmatrix} = \begin{bmatrix} I & 0 & 0 \\ 0 & 0 & I \end{bmatrix}.$$

If A is square and invertible, the first row requires

$$X = A^{-1}, \qquad Y = 0, \qquad Z = -A^{-1}B.$$

Each formula follows by multiplying its block equation on the left by A^{-1}. If A is rectangular with a right inverse X, one possible solution is instead $Y = 0, Z = -XB$; $AX = I$ checks it. An A without a right inverse cannot satisfy the first equation.

Exercise 2.4.9: Eliminate two blocks below a leading block

The second block row starts with $A_{21}B_{11} + B_{21}$. Setting it to zero and multiplying on the right by B_{11}^{-1} gives $A_{21} = -B_{21}B_{11}^{-1}$. Likewise $A_{31} = -B_{31}B_{11}^{-1}$. The second block in the new second row is consequently

$$C_{22} = A_{21}B_{12} + B_{22} = B_{22} - B_{21}B_{11}^{-1}B_{12}.$$

For completeness $C_{11} = B_{11}$, $C_{12} = B_{12}$, and $C_{32} = B_{32} - B_{31}B_{11}^{-1}B_{12}$. The middle inverse cannot be moved past its neighboring factors.

Exercise 2.4.10: Invert three block levels in order

Multiply the original matrix by the proposed inverse. Its lower blocks must satisfy

$$A + P = 0, \qquad D + R = 0, \qquad B + DP + Q = 0.$$

Thus $P = -A$, $R = -D$, and $Q = DA - B$. In the reverse product the lower-left block is $Q + RA + B = (DA - B) - DA + B = 0$, while the other off-diagonal blocks also vanish. Hence both products are the identity.

Conclusion. $P = -A$, $Q = DA - B$, and $R = -D$.

Exercise 2.4.11: Check conformability rather than only the number of blocks

(a) **True.** Corresponding blocks have equal sizes, so adding entries gives $[A_1 + B_1 \;\; A_2 + B_2]$.

(b) **False.** The drawn partitions do not specify the actual sizes. The number of columns of each A_{ij} must equal the number of rows of the corresponding B_j. For example, take all A_{ij} scalar but each B_j a 2×1 column. The displayed block pattern exists but multiplication is not conformable.

Exercise 2.4.12: Dimensions and transposes of partitions

(a) **False.** A has size $2n \times n$ and B has size $n \times 2n$. Both BA $(n \times n)$ and AB $(2n \times 2n)$ are defined. The blocks of AB are A_iB_j.

(b) **False.** Transposition interchanges block positions as well as transposing each block: $A^T = \begin{bmatrix} P^T & R^T \\ Q^T & S^T \end{bmatrix}$. The stated version puts the off-diagonal blocks in the wrong places.

Exercise 2.4.13: Invertibility of a block diagonal matrix

If B and C are invertible, block multiplication shows that $\operatorname{diag}(B^{-1}, C^{-1})$ is a two-sided inverse. Conversely suppose $\operatorname{diag}(B, C)$ is invertible. If $Bu = 0$, then the block vector $(u, 0)^T$ is in the kernel of the whole matrix; injectivity forces $u = 0$. Since B is square, the Invertible Matrix Theorem makes B invertible. The same argument with $(0, v)^T$ proves C invertible.

Conclusion. The equivalence holds in both directions.

Exercise 2.4.14: Invertibility of a block upper triangular matrix

Write $A = \begin{bmatrix} A_{11} & A_{12} \\ 0 & A_{22} \end{bmatrix}$ with square diagonal blocks. If they are invertible, the two-sided inverse is

$$\begin{bmatrix} A_{11}^{-1} & -A_{11}^{-1}A_{12}A_{22}^{-1} \\ 0 & A_{22}^{-1} \end{bmatrix}.$$

The upper-right block of the product in the original order is $-A_{12}A_{22}^{-1} + A_{12}A_{22}^{-1} = 0$; in the other order it is $A_{11}^{-1}A_{12} - A_{11}^{-1}A_{12} = 0$. For necessity, invertibility of A first forces A_{11} invertible: a nonzero u with $A_{11}u = 0$ would give kernel vector $(u, 0)^T$. If $A_{22}v = 0$ with $v \neq 0$, put $u = -A_{11}^{-1}A_{12}v$. Then $A(u, v)^T = 0$, another contradiction. Thus A_{22} also has trivial kernel and is invertible.

Exercise 2.4.15: Update an accumulated outer-product sum

The column–row rule gives

$$G_k = X_kX_k^T = \sum_{j=1}^{k} x_jx_j^T.$$

Appending the new column yields

$$G_{k+1} = [X_k \;\; x_{k+1}] \begin{bmatrix} X_k^T \\ x_{k+1}^T \end{bmatrix} = G_k + x_{k+1}x_{k+1}^T.$$

Only one new outer product is needed, followed by entrywise addition into the stored G_k. If each x_j has m entries, the new product uses m^2 scalar products of entries, rather than recomputing $k+1$ outer products. The already processed vectors need not be revisited.

Exercise 2.4.16: Derive the Schur-complement factorization

Multiply the last two factors first:

$$\begin{bmatrix} A_{11} & 0 \\ 0 & S \end{bmatrix} \begin{bmatrix} I & Y \\ 0 & I \end{bmatrix} = \begin{bmatrix} A_{11} & A_{11}Y \\ 0 & S \end{bmatrix}.$$

Multiplying the left factor then gives $\begin{bmatrix} A_{11} & A_{11}Y \\ XA_{11} & XA_{11}Y + S \end{bmatrix}$. Choose $Y = A_{11}^{-1}A_{12}$ and $X = A_{21}A_{11}^{-1}$. The lower-right block becomes $A_{21}A_{11}^{-1}A_{12} + S = A_{22}$ by the definition of S. All four blocks now agree with the required matrix.

Exercise 2.4.17: Invertibility passes to the Schur complement

The outside factors from Exercise 2.4.16 are invertible: reverse X or Y in their off-diagonal block, just as in Practice Problem 1. If $A = L\,\mathrm{diag}(A_{11}, S)U$ is invertible, then $\mathrm{diag}(A_{11}, S) = L^{-1}AU^{-1}$ is invertible. Exercise 2.4.13 implies S is invertible. Inverting the factors in reverse order also gives the explicit formula

$$A^{-1} = \begin{bmatrix} A_{11}^{-1} + A_{11}^{-1}A_{12}S^{-1}A_{21}A_{11}^{-1} & -A_{11}^{-1}A_{12}S^{-1} \\ -S^{-1}A_{21}A_{11}^{-1} & S^{-1} \end{bmatrix}.$$

This expression follows by multiplying U^{-1}, the block diagonal inverse, and L^{-1}, in that order.

Exercise 2.4.18: Append one data column to a cross-product matrix

For $W = [X\ x_0]$, transposition and block multiplication give

$$W^TW = \begin{bmatrix} X^TX & X^Tx_0 \\ x_0^TX & x_0^Tx_0 \end{bmatrix}.$$

Its Schur complement of the invertible upper-left block is the scalar

$$x_0^Tx_0 - x_0^TX(X^TX)^{-1}X^Tx_0 = x_0^T[I - X(X^TX)^{-1}X^T]x_0 = x_0^TMx_0.$$

The asserted reciprocal entry requires this scalar to be nonzero; otherwise W^TW is singular, and an inverse entry is not defined. The question's requested identity itself holds regardless of whether the scalar vanishes.

Exercise 2.4.19: Eliminate the state vector to obtain the transfer function

Put $H = A - sI_n$, assumed invertible. The top block equation is $Hx + Bu = 0$, so $x = -H^{-1}Bu$. Insert this into the bottom equation $Cx + u = y$ to obtain

$$[I_m - CH^{-1}B]u = y.$$

Thus $W(s) = I_m - C(A - sI_n)^{-1}B$. It is exactly the Schur complement of $A - sI_n$ in the original block system matrix. Its size is $m \times m$, so it acts on the input vector $u \in \mathbb{R}^m$.

Exercise 2.4.20: The inverse transfer function is another Schur complement

The Schur complement of $K = A - BC - sI_n$ in the displayed matrix is

$$I_m - (-C)K^{-1}B = I_m + C(A - BC - sI_n)^{-1}B.$$

This is the requested $W(s)^{-1}$ when the stipulated inverses exist. To check the inverse claim, write $K = H - BC$, $H = A - sI_n$. From $K = H - BC$, multiply $KK^{-1}B = B$ on the left by H^{-1} and rearrange to get the identity

$$K^{-1}B = H^{-1}B + H^{-1}BCK^{-1}B.$$

Multiplying it by C shows $CK^{-1}B = CH^{-1}B + CH^{-1}BCK^{-1}B$. Expanding $(I - CH^{-1}B)(I + CK^{-1}B)$ now cancels these three terms and gives I, confirming the formula. The invertibility of H and W also forces K invertible: if $Kv = 0$, then $v = H^{-1}BCv$ and $W(Cv) = 0$; hence $Cv = 0$ and then $v = 0$. The square-matrix theorem completes this check.

Exercise 2.4.21: Verify an involution using blocks

(a) Multiply the 2×2 matrix.

$$\begin{bmatrix} 1 & 0 \\ 2 & -1 \end{bmatrix}^2 = \begin{bmatrix} 1 & 0 \\ 2-2 & 1 \end{bmatrix} = I_2.$$

(b) Recognize the four-dimensional block structure.

For this same A, the given M is $\begin{bmatrix} A & 0 \\ I_2 & -A \end{bmatrix}$. Consequently

$$M^2 = \begin{bmatrix} A^2 & 0 \\ A - A & A^2 \end{bmatrix} = I_4.$$

The lower-left cancellation is essential; squaring only the diagonal blocks would not verify the entire product.

Exercise 2.4.22: Construct a six-dimensional involution

Choose $A = B = I_2$, $D = -I_2$, and $C = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \neq 0$. Then

$$M = \begin{bmatrix} I_2 & 0 & 0 \\ 0 & I_2 & 0 \\ C & 0 & -I_2 \end{bmatrix}.$$

It has size 6×6 as required. Its square has identity diagonal blocks and the only possibly nonzero off-diagonal block is $CA + DC = C - C = 0$. Thus $M^2 = I_6$. This construction works with any nonzero 2×2 choice for C.

Exercise 2.4.23: Induct on the dimension of lower triangular matrices

For dimension one the product of two scalars is lower triangular. Assume products of $k \times k$ lower triangular matrices are lower triangular. Partition two matrices of dimension $k+1$ as

$$A_1 = \begin{bmatrix} a & 0^T \\ v & A \end{bmatrix}, \qquad B_1 = \begin{bmatrix} b & 0^T \\ w & B \end{bmatrix}.$$

Their product is

$$A_1 B_1 = \begin{bmatrix} ab & 0^T \\ vb + Aw & AB \end{bmatrix}.$$

The top-right block is zero and AB is lower triangular by the induction hypothesis. These facts account for every entry above the main diagonal. Thus the product in dimension $k+1$ is lower triangular, completing the induction.

Exercise 2.4.24: Invert cumulative sums by successive differences

Let A_n be the lower triangular matrix of ones and B_n the matrix with ones on the diagonal and -1 immediately below it. For $n = 2$, direct multiplication of $\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$ and $\begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$ in either order gives I_2. Suppose $A_k B_k = B_k A_k = I_k$. With $\mathbf{1}$ the k-entry ones vector and $e_1 = (1, 0, \ldots, 0)^T$, partition

$$A_{k+1} = \begin{bmatrix} 1 & 0^T \\ \mathbf{1} & A_k \end{bmatrix}, \quad B_{k+1} = \begin{bmatrix} 1 & 0^T \\ -e_1 & B_k \end{bmatrix}.$$

The lower-left block of $A_{k+1}B_{k+1}$ is $\mathbf{1} - A_k e_1 = 0$, since $A_k e_1$ is the first column, all ones. The lower-left block of the reverse product is $-e_1 + B_k \mathbf{1} = 0$: successive differences of ones are $(1, 0, \ldots, 0)^T$. The diagonal blocks are identities by induction and the upper-right blocks are zero. Both products are I_{k+1}, which proves the assertion for every $n \geq 2$.

Exercise 2.4.25: Invert the three diagonal blocks separately

The given matrix is block diagonal with blocks $B = \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}$, $[2]$, and $C = \begin{bmatrix} 7 & 8 \\ 5 & 6 \end{bmatrix}$. The 2×2 formula gives

$$B^{-1} = \begin{bmatrix} -5 & 2 \\ 3 & -1 \end{bmatrix}, \qquad C^{-1} = \begin{bmatrix} 3 & -4 \\ -5/2 & 7/2 \end{bmatrix}.$$

Thus, without row reduction,

$$A^{-1} = \begin{bmatrix} -5 & 2 & 0 & 0 & 0 \\ 3 & -1 & 0 & 0 & 0 \\ 0 & 0 & 1/2 & 0 & 0 \\ 0 & 0 & 0 & 3 & -4 \\ 0 & 0 & 0 & -5/2 & 7/2 \end{bmatrix}.$$

Each product block is an identity, so the whole product is I_5.

Exercise 2.4.26: Concrete commands for extracting and inserting blocks

Using Python/NumPy's zero-based, end-exclusive slices:

(a) `A[4:10, 14:20]` displays rows 5–10 and columns 15–20. The result has size 6×6.

(b) `A[4:9, 9:19] = B` inserts the 5×10 block in rows 5–9 and columns 10–19. The first inserted entry is at row 5, column 10, as required.

(c) `scipy.linalg.block_diag(A, A.T)` constructs the desired 50×50 block diagonal matrix. Alternatively initialize a 50×50 zero array, assign A to `C[:20, :30]` and A^T to `C[20:, 30:]`. The off-diagonal blocks then already have the necessary zero entries.

These are the requested operations, not a request for unspecified data. Their output sizes follow from the slice endpoints and the given sizes.

Exercise 2.4.27: Work entirely with blocks below the size limit

Keep the large matrices on disk or as separately stored blocks; assembling a full 50×50 array would itself violate the limit.

(a) Partition both matrices into four 25×25 blocks. Compute each sum block $C_{ij} = A_{ij} + B_{ij}$ separately and store the four results as the sum's block representation.

(b) With that same partition compute $C_{ij} = A_{i1}B_{1j} + A_{i2}B_{2j}$ for $i, j \in \{1, 2\}$. Every individual multiplication and addition is 25×25, well within the 32×32 bound.

(c) Use the stipulated $20 + 30$ partition, with $b = (b_1, b_2)^T$ and $x = (x_1, x_2)^T$. Since $A_{12} = 0$, first solve the 20×20 system $A_{11}x_1 = b_1$. Compute $r = b_2 - A_{21}x_1$ (A_{21} is 30×20), then solve $A_{22}x_2 = r$, a 30×30 system. Both diagonal matrices are invertible, so both solves have unique answers. Store x_1, x_2 separately as the solution; no explicit inverse or overlarge matrix is needed.